

IBM i Modernization

Modernizing RPG Applications, Green Screens, and Data Access Patterns

IBM Bob modernizes IBM i applications by transforming Fixed-Format RPG, Green Screen transformation and data access while preserving business continuity.

80%

of IBM i business logic is embedded in RPG, CL and DDS applications with limited documentation

70% faster

Fixed-Format RPG to Free-Format RPG modernization

60% faster

Green screen modernization and API enablement

80% faster

Application understanding and documentation



Business Challenges

- Critical business logic resides in senior RPG developers and undocumented legacy applications, creating operational risk.
- Retirement of skilled IBM i developers is creating a growing knowledge and skills gap.
- Organizations need modern user experiences, APIs, and cloud integration while maintaining business continuity.
- Fixed-format RPG, green-screen applications, and Record Level Access (RLA) increase maintenance costs and limit agility.
- Traditional modernization efforts require significant manual analysis, documentation, and migration effort.



Solution Highlights

- Fixed-format RPG → Free-format RPG conversion.
- Green Screen (5250) → Modern Web UI and API generation.
- RLA → Embedded SQL modernization and optimization.
- Direct IBM i connectivity for read, write, compile, test, and deploy.
- Pre-built IBM i skills for understanding, refactoring, documenting, testing, and transforming applications.



Business Value

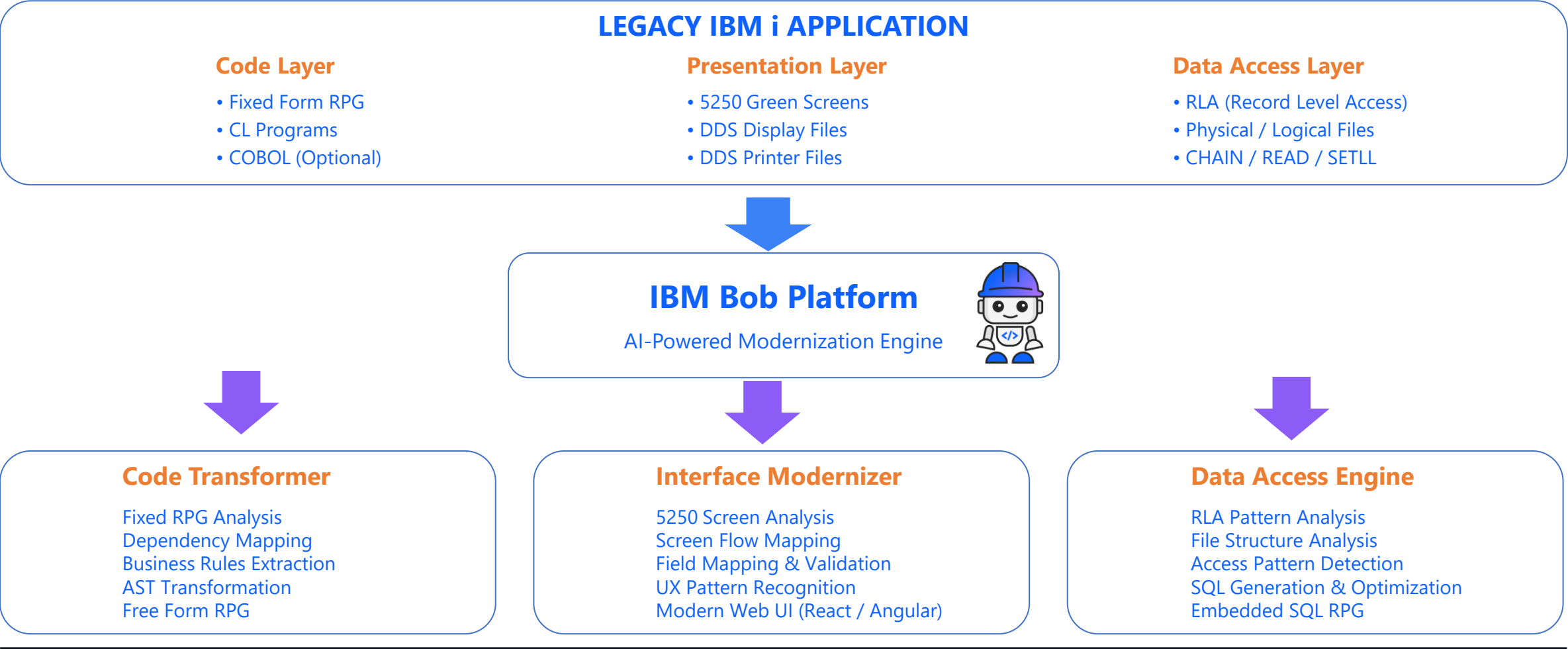
- Capture and document decades of business logic embedded within RPG, CL, DDS, and COBOL applications.
- Accelerate onboarding and developer productivity.
- Modernize fixed-format RPG, green screens, and legacy data access patterns to support API-first and digital transformation initiatives.
- Lower modernization costs and timelines.
- Reduce modernization risk through AI-assisted analysis and testing.
- Shift team capacity from maintenance to innovation.

CONTRIBUTORS

[[Ranjan Kumar Jena](#)] · AI Engineer

Reference Architecture

Modernizing RPG Applications, Green Screens and Data Access Patterns



Reduce technical debt, improve developer productivity, and extend the value of mission-critical IBM i applications